



AI-Driven Systematic Reviews: Transformer-Based Approaches to Text Classification

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Abstract

This paper explores the application of transformer-based models to automate the systematic literature review process, a traditionally labor-intensive task. Utilizing advanced NLP models from Hugging Face AutoTrain, a series of experiments were conducted on three manually annotated datasets: Mental Health (MH), Explainable Artificial Intelligence (EXAI), and Technology in Learning (LEARN). The experiments involved the fine-tuning of various transformer-based models and the exploration of ensemble approaches to enhance performance. The evaluation was performed using standard metrics, including accuracy, recall, and the F3 measure. Fine-tuned transformers demonstrated high accuracy and recall, particularly on balanced datasets, while ensemble models provided stable classification results by leveraging multiple transformers. Despite promising results, challenges such as high computational costs and the need for extensive hyperparameter tuning remain. This research demonstrates the potential of AI in supporting systematic literature reviews, reducing the workload while maintaining high accuracy.

CCS Concepts

• Information systems; • Information retrieval; • Specialized information retrieval;

Keywords

Transformer models, Ensemble learning, Text classification, Systematic literature review, Natural language processing (NLP)

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1 Introduction

Scientific papers strive to build upon, verify, or challenge prior research. A systematic literature review is an important step in writing a scientific paper, as it provides an overview of the advancements already achieved in a specific field, deepens and broadens the researcher's understanding and knowledge. The literature review also provides a comparison and, sometimes, an inspiration for a paper or a project. However, this process often involves a large number of papers and is therefore very time-consuming for the individuals doing the research. This problem can be solved by increasing the size of the team doing the review, but it can also lead to workflow disruption or inconsistency in the review process.

By using artificial intelligence (AI), machine learning (ML), and deep learning (DL) models, in combination with natural language processing (NLP) techniques, the process can be automated to a certain degree, with the tools and models doing some of the work, allowing researchers to become more efficient [3].

According to the PRISMA framework [13, 33], checklist items include the specification of inclusion and exclusion criteria (eligibility criteria). Since the PRISMA methodology classifies the papers as included or excluded, the classification process can be considered as a binary classification task performed using the ML model.

Motivated by the success of large language models and transformers in NLP tasks and text classification, this paper presents the use of transformers, state-of-the-art machine learning models, to automate aspects of the traditionally labor-intensive systematic literature review process.

For this purpose, we conducted a series of experiments involving the fine-tuning of various transformer-based models from Hugging Face AutoTrain on three manually annotated datasets of research papers on specific topics: MH, EXAI, and LEARN for the task of scientific text classification. We performed extensive hyperparameter tuning to achieve optimal performance. Additionally, we experimented with ensemble models that combine multiple transformers.

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The models were evaluated using several standard metrics, with a particular focus on accuracy, recall, and the F3 score.

The contribution of this paper is an overview of the performance of various transformer-based models and their ensembles for classification tasks in the context of systematic literature reviews.

This paper is organized as follows: Section 2 gives a theoretical background to the literature review process and used AI models and tools, section 3 presents related work to the topic of the research, section 4 introduces used resources and methodology which produced results displayed in section 5. Section 6 provides a conclusion to the paper.

2 Theoretical Background

2.1 Literature Review Process

The literature review process is an essential aspect of scientific research and serves as the basis for extending previous studies [21]. This process provides researchers with an insight into the current state of knowledge in a field and helps to identify areas where further investigation is required.

A systematic literature review (SLR) is a structured approach to summarizing existing research relevant to a particular hypothesis or research question [31]. The primary goal of an SLR is to summarize the existing literature with clarity and transparency and to ensure that the results are both comprehensive and unbiased. To achieve this, various methods have been developed to standardize the process. One widely recognized method is the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) method, which outlines 27 critical steps to improve the transparency and reproducibility of literature reviews [14].

The PRISMA method provides a detailed framework that researchers can follow to ensure systematic data collection and analysis. This approach includes defining the research question, establishing inclusion and exclusion criteria, conducting a comprehensive search of relevant databases, and systematically reviewing and selecting studies. Each step aims to reduce bias and improve the reliability of the results.

However, conducting an SLR is resource intensive. Manually searching, selecting and analyzing a large number of manuscripts often requires a lot of time and effort. The process can also require a large team to maintain accuracy and reduce the risk of errors due to reviewer fatigue. If the workload is unevenly distributed among too few individuals, the potential for error increases, impacting both the quality of the review and the associated costs.

PRISMA not only provides a methodical roadmap, but also a visual representation template to illustrate the flow of information through the various stages of the review. This graphical representation helps to make transparent the number of studies assessed, included or excluded at the different stages of the review process. Such visualization supports clearer communication of the systematic approach and highlights the thoroughness of the review.

Despite its structured nature, the systematic review process presents challenges that researchers must address. These include providing sufficient resources and maintaining rigorous standards throughout the process. Overcoming these challenges is critical to producing high quality, reliable literature reviews that contribute meaningfully to the field of study.

2.2 NLP and LLMs

Over the last decade, significant advancements in NLP capabilities have been achieved, with improvements in performance, precision, adaptability, and scalability, allowing models to process vast amounts of text data efficiently and apply sophisticated understanding to complex language tasks [10]. Earlier NLP systems relied on handcrafted features and predefined rules, which limited their adaptability and effectiveness in complex NLP tasks. By contrast, modern NLP models, based on LLMs and supported by deep learning, automatically extract complex features from text, allowing better language representation and understanding.

The main advantage of LLMs lies in their ability to capture subtle patterns in language, including syntactic structures, complex semantic relationships, domain-specific terminology, and idiomatic expressions. LLMs undergo extensive pretraining on vast datasets, enabling them to capture deep language patterns and context, which supports robust generalization across different tasks and domains. Consequently, pre-trained language models require minimal fine-tuning for adaptation to specific applications.

These developments have allowed NLP systems to outperform classical NLP algorithms achieving state-of-the-art results across various NLP tasks that were previously limited by less adaptable approaches [17]. These tasks span from basic text parsing and information retrieval to more complex, such as text classification, sentiment analysis, content summarization, question answering, translation, etc.

Building on the success of LLMs, much of their recent improvement is due to the transformer architecture introduced by Vaswani et al. [29] in 2017. Unlike earlier models, such as recurrent neural networks (RNNs) and long short-term memory networks (LSTMs), which processed text sequentially and struggled with long-range dependencies, transformers apply a self-attention mechanism that captures relationships between all words in a sequence simultaneously. This innovation allows transformers to handle longer contexts more effectively, making them particularly suited to processing complex and often lengthy texts across various domains. As a result, transformers overcome the limitations of RNNs, which can suffer from information loss over long sequences and require more extensive training time due to their sequential nature. This improvement has led to large pre-trained language models based on the transformer architecture, such as BERT [5] and GPT [22], becoming central to recent progress in NLP. BERT stands out as one of the most influential transformer-based models, designed specifically to capture context from both directions in a sentence. It utilizes a bidirectional approach to pretraining, allowing it to develop a deeper understanding of linguistic patterns and relationships between words in context.

In general, transformer-based models consistently outperform traditional approaches in various tasks related to text classification [16]. Specifically, when applied to the task of scientific paper classification, transformers offer a few distinct advantages [2]. They can effectively distinguish between specialized topics and domains by capturing complex, domain-specific terminology and contextual cues that are often difficult for traditional models to interpret [7]. Compared to traditional deep neural network models, transformers offer distinct advantages in accuracy and efficiency.

Overall, in the context of the literature review process, NLP, LLMs and specifically transformer-based LLMs can be a powerful tool for automating the tasks of searching, selecting, and analyzing large volumes of manuscripts [9]. NLP techniques enable efficient extraction, classification, and synthesis of information from extensive research literature, reducing the time and effort required for manual processing [18]. Furthermore, through automated topic identification, citation analysis, and document clustering, NLP may help in the organization and categorization of scientific papers, ensuring consistency and accuracy.

3 Related Work

This section provides a brief overview of related work on the application of NLP methods for literature review tasks, covering both earlier studies that used traditional NLP methods and recent research applying LLMs to these tasks.

Earlier studies widely explored the application of NLP techniques for various tasks in the literature review process. Traditional NLP methods, such as text mining, have been extensively used to support systematic reviews, especially in identifying relevant studies and extracting key information. For instance, O'Mara-Eves et al. [19] conducted a systematic review of text mining methods for identifying relevant studies to include in systematic reviews. Another study describes the application of text mining to four tasks, namely, document clustering, document classification, text summarization, and automatic term recognition in the task of identification of relevant studies within the literature review process [27]. Advanced text mining approaches for automatic literature review have also been explored in [12], which proposed a framework with three self-defined semantics-based ranking metrics: keyword relevance, indexed-term relevance, and topic relevance. These early studies and applications have shown that NLP methods have the potential to automate literature review tasks. With the recent advent of generative AI and LLMs, NLP methods have become even more powerful and adaptable for such tasks.

In a recent study [28], the authors present a survey of AI techniques designed to assist researchers in performing systematic analyses of scientific literature. They describe the supported tasks, types of algorithms, and available tools proposed in 34 primary studies offering insights into the evolution of automated literature review processes. Many recent studies have explored the use of NLP techniques, such as text classification and topic modeling, to automate parts of the review process, thereby improving its speed and consistency. For instance, Marshall and Wallace [15] explore text classification and other machine learning approaches for research synthesis and literature review automation, while [30] examine topic modelling as a solution for analyzing scientific paper content by introducing an LDA-based tool, LDAShiny.

Recent work highlights the potential of transformer-based models in addressing the challenges of scientific text analysis, such as processing long documents [8], handling domain-specific terminology, and overcoming the lack of large-scale labelled datasets [4]. One approach introduces Hierarchical Transformers, which extends BERT's architecture by processing long sequences, achieving innovative classification capabilities for scientific texts [20]. Another method presents a transformer-based model for scientific

texts classification, designed as a parameter-efficient solution with a lightweight multiheaded attention encoder and hybrid embeddings [1].

There are several LLMs based on BERT architecture, that are being specifically developed and trained for the representation of scientific texts and various NLP tasks related to scientific literature, such as the classification of scientific texts. Among these, SciBERT [3], introduced by Beltagy et al., is a BERT-based model further trained on a large corpus of scientific texts. Similarly, other domain-specific BERT variants trained on scientific texts have been developed, such as BioBERT [11] for the biomedical domain and SocioBERT, a pre-trained model based on abstracts from SSCI journals, for the social sciences [24]. Furthermore, in [2], the authors propose a novel parameter-efficient transformer-based model called sBERT. This model was developed and optimised for the task of classifying scientific literature. The results of extensive experiments conducted on datasets from Web of Science, ArXiv, Nature, Springer, and Wiley show that sBERT performs better than traditional models in terms of accuracy and efficiency.

Previous research indicates the significant potential of NLP in tasks related to the literature review process. In particular, transformer-based models and variants of the BERT model seem to be effective tools for the classification of scientific texts. Although many studies explore LLMs and transformer-based models for text classification, there is still a limited number of papers investigating the performance of these models specifically for scientific text classification, including the use of ensembles of transformer-based models. With this in mind, in this study, we explore various transformer-based models and their ensembles for tasks related to literature review.

4 Methods and Tools

The methodology of the research is based on the use of transformer models and the Hugging Face AutoTrain tool [34] described in subsection 4.2, along with other resources used in the research. The AutoTrain tool is used to fine-tune models already trained on large text corpora to improve the metrics obtained in text classification tasks. Features of the papers used for classification were their titles and abstracts. To fine-tune and evaluate the performance of the model, three existing manually created datasets for literature reviews were used and had to be pre-processed to make them suitable for model input, described in 4.1. Subsection 4.3 explains the metrics used to evaluate classification performance. A detailed description of the steps taken to fine-tune models and find the optimal hyperparameters can be found in subsection 4.4.

4.1 Datasets

Three different datasets were used to train and evaluate different language models based on the transformer architecture. The datasets are labeled as MH, EXAI and LEARN and are created as part of the separate manual literature reviews.

The MH dataset focuses on finding papers relevant for the technology in the treatment of mental illness. The inclusion criteria are designed to include papers on video game development for

mental health. The dataset contains 178 examples of titles and abstracts, with both relevant and irrelevant papers represented with 89 examples each. The creation of the dataset is described in [3].

The EXAI dataset provides an insight into the field of explainable artificial intelligence. The EXAI dataset consists of 175 examples, with 88 positively classified and 87 negatively classified examples. A more detailed description of the dataset and the methods used to create it is available in [3].

The LEARN dataset consists of papers that deal with the use of modern technologies in the personalized learning process. This dataset is also mentioned in [6]. The dataset is substantially larger than the EXAI and MH datasets and consists of 3149 examples. In contrast to the EXAI and MH datasets, the LEARN dataset is not balanced and contains 2962 negatively labeled examples and only 187 positive examples. Due to the unbalanced labels, changes had to be made to the training and test datasets to achieve the best possible results. As a result, this study considers a dataset variant with a sampled and balanced training set, labeled LEARN_{SB}, which stands for LEARN-sampled-balanced. Experiments have shown that unbalanced training sets resulted in non-acceptable performance, so only the LEARN_{SB} variant was considered during the analysis of the results.

4.1.1 Dataset Preprocessing. The three datasets had to be preprocessed to be appropriate for fine-tuning the model with AutoTrain. Common NLP techniques were used, such as decontraction and the removal of numbers and special characters. Stop word removal may not be appropriate for some NLP applications where sentence structure is important, but in this case stop words were removed, as many models have limited input size and cannot accommodate all the text required for specific examples. Lemmatization was used as opposed to stemming, as the models' results were less favorable when stemming was used for almost all examples tested. Since the AutoTrain user interface only accepts one column from which it determines the class, it was necessary to concatenate the abstract and title of the paper into a single text entry.

4.2 Training Resources

4.2.1 Hugging Face. Hugging Face [32] is an open-source platform that aims to provide accessible resources for machine learning and enable the use of complex machine and deep learning models, such as transformers. Hugging Face provides ability to use and train models, such as the Transformers library [35] for Python and Hugging Face Hub. This platform is currently extremely popular for all types of machine learning tasks, not just the text classification used in this research. Its user-friendly and easy-to-use libraries make it an appealing platform for training, evaluating, and deploying models.

4.2.2 AutoTrain. AutoTrain is a feature of Hugging Face, that enables models integrated into the Hugging Face ecosystem to be trained or fine-tuned and evaluated independently of the model's actual architecture [31]. AutoTrain training on Hugging Face Spaces and the local command line interface (CLI) and user interface (UI) were used for fine-tuning all tested transformers [33]. Fine-tuning enabled better results than training from scratch, as the models available on Hugging Face and AutoTrain were already trained on

large text datasets and therefore show good performance on NLP tasks in combination with the transformer architecture.

4.2.3 Transformers. In this research different models were used with the transformer architecture, from widely used BERT-based models to less known models that still perform great, such as all-mpnet-base-v2 and splade-cocondenser-ensembledistil. Models trained for specific topic corpora were excluded, e.g. models from the financial or medical domains. A list of the models used in this research is provided in Table 1.

4.2.4 Learning. Ensemble learning utilizes the power of more than one transformer to obtain the best possible results. The technique used is "max pooling", a type of majority voting in which the class assigned to an example by a model is the most frequent class among all models contained in an ensemble. The ensemble learning method could improve results on datasets whose classification results are inadequate. However, it could also lead to correct classification results being out-voted by incorrect classifications of other models in an ensemble, leading to worse results. Ensembles were evaluated by combining transformers in different ensembles using the Ensemble Transformers library [26].

4.3 Metrics

The following metrics were used when testing the model's accuracy, AUC score, precision and recall, along with F1, F2 and F3 [25].

Accuracy, one of the most used metrics in evaluating machine learning models, measures how often the model correctly classifies the paper, while the AUC score indicates the overall quality of a classification model. Precision indicates how many truly relevant papers are among the ones the model has classified as relevant. Recall, one of the more important metrics in the context of this research, indicates the percentage of relevant papers that the model has classified as relevant. The importance of recall stems from the nature of the literature review process, in which a person reviews papers that the model has classified as relevant. If a model classifies an irrelevant paper as relevant (a false positive), the person doing the literature review can correct this mistake. On the other hand, if a model discards a relevant paper (a false negative classification option), the paper is not included in the model output and is discarded. Thus, to prevent relevant papers from being overlooked, the goal is to maximize the true-positive rate (TPR) and improve recall as much as possible while keeping accuracy and AUC as high as possible to reduce the workload of human raters. With that in mind, the F-beta score was included, where betas are 1, 2 and 3. F-beta represents a weighted harmonic mean of precision and recall, as shown in formula below [23].

$$F_{\beta} = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{(\beta^2 \cdot \text{precision}) + \text{recall}}$$

F1 serves as a metric to show the quality of the precision and recall metrics as one metric. F2 and F3 are used to rank the models based on a criterion that prioritizes recall over precision. The rationale behind this approach is the importance of classifying as many relevant results as possible as positive, even at the expense of an increased number of false positives. We consider this approach particularly valuable in the context of identifying relevant scientific papers.

Table 1: Hugging Face models for the text classification task selected and fine-tuned in this research for scientific paper classification.

Model name	Hugging Face model name
ALBERT Base v2	albert/albert-base-v2
DistilBERT base model (uncased)	distilbert/distilbert-base-uncased
DistilRoBERTa base	distilbert/distilroberta-base
RoBERTa base model	FacebookAI/roberta-base
XLM-RoBERTa (base-sized model)	FacebookAI/xlm-roberta-base
BERT base model (uncased)	google-bert/bert-base-uncased
DeBERTaV3	microsoft/deberta-v3-base
SPLADE CoCondenser EnsembleDistil	naver/splade-cocondenser-ensembledistil
SPLADE CoCondenser SelfDistil	naver/splade-cocondenser-selfdistil
all-distilroberta-v1	sentence-transformers/all-distilroberta-v1
all-mpnet-base-v2	sentence-transformers/all-mpnet-base-v2
multi-qa-distilbert-cos-v1	sentence-transformers/multi-qa-distilbert-cos-v1
multi-qa-mpnet-base-cos-v1	sentence-transformers/multi-qa-mpnet-base-cos-v1
multi-qa-mpnet-base-dot-v1	sentence-transformers/multi-qa-mpnet-base-dot-v1

Table 2: AutoTrain Hyperparameters.

Hyperparameter	Value
Mixed precision	fp16
Optimizer	adamw_torch
Scheduler	linear
Batch size	8
Epochs	3
Gradient accumulation	1
Learning rate	0.00005
Max sequence length	512

4.4 Evaluation Process

4.4.1 Fine-tuning and Evaluating Transformers. The most popular transformers for the text classification task provided in the Hugging Face AutoTrain were applied and evaluated in the scientific text classification task. The selected models (listed in 4.2.3) were trained with a set of fixed hyperparameters which are listed in Table 2.

All hyperparameters shown are the default values, except for the maximum sequence length. The maximum sequence length was set to 512, as this is the limit of the models used. Also, most of the examples in the datasets were larger than the default value of 128, so many of the abstracts were not fully included.

After the initial tests, some of the models were discarded due to poor results. Those that achieved better performance were subjected to further testing, varying different values of the hyperparameters to achieve better classification results. Optimal hyperparameters differed between models due to different architecture, and each model had to be trained by varying the hyperparameter values to determine the best performance. In general, the only pattern found among the models and their hyperparameters was that increasing the epoch size led to better results in many cases. However, too high an epoch number led to overfitting of the model and degradation of the model’s classification capabilities. Training hyperparameters used in fine-tuning models whose performance is presented

in Table 3 as part of the result are as follows: Deberta-v3-base and all-mpnet-base-v2 were trained with the hyperparameters shown in Table 2. There were some differences in the other models: splade-cocondenser-ensembledistil had 5 epochs set, bert-base-uncased had a batch size of 5, multi-qa-mpnet-base-dot-v1 had a batch size of 7, 6 epochs and a learning rate of 0.0006 and multi-qa-distilbert-cos-v1s had 4 epochs and a learning rate of 0.0003. After the optimal hyperparameters were determined, the models were evaluated on all three datasets using metrics explained in subsection 4.3. For EXAI and MH, the train sets contained 140 examples, while the LEARN dataset’s train set contained 200 examples of titles and abstracts, the rest of the respective datasets formed the test set.

4.4.2 Evaluating Ensembles. Having already fine-tuned the models with the hyperparameters that achieved the best results, ensembles were formed by combining them with the Ensembles Transformers Python library [26]. Sets of 3, 4 and 5 transformers were formed by selecting models from a pool of models with the best overall performance for all three datasets. Additionally, transformers that did not perform well in all three datasets, but only in one or two datasets, were included to improve the accuracy and recall metrics of the datasets. Different combinations of these models were then ensembled and tested with the same training and test sets as the individual models.

Table 3: Selected text classification models' evaluation results on the literature review task.

Base model	Dataset	Accuracy	AUC score	Precision	Recall	F1	F2	F3
all-mpnet-base-v2	EXAI	0.886	0.902	0.899	0.899	0.899	0.899	0.899
	MH	0.917	0.948	0.895	0.944	0.919	0.934	0.939
	LEARNSB	0.833	0.936	0.145	0.954	0.252	0.451	0.621
	average	0.879	0.929	0.646	0.932	0.690	0.761	0.820
splade-cocondenser-ensembledistil	EXAI	0.943	0.951	0.944	0.944	0.944	0.944	0.944
	MH	0.889	0.932	0.850	0.940	0.893	0.921	0.930
	LEARNSB	0.826	0.921	0.142	0.978	0.248	0.449	0.616
	average	0.886	0.935	0.645	0.954	0.695	0.771	0.830
bert-base-uncased	EXAI	0.914	0.958	0.941	0.889	0.914	0.899	0.894
	MH	0.861	0.923	0.842	0.889	0.865	0.879	0.884
	LEARNSB	0.795	0.937	0.126	1,000	0.224	0.419	0.590
	average	0.857	0.939	0.636	0.926	0.668	0.732	0.789
multi-qa-mpnet-base-dot-v1	EXAI	0.914	0.905	0.941	0.889	0.914	0.899	0.894
	MH	0.889	0.917	0.889	0.889	0.889	0.889	0.889
	LEARNSB	0.838	0.938	0.147	0.943	0.254	0.453	0.612
	average	0.880	0.920	0.659	0.907	0.686	0.747	0.798
multi-qa-distilbert-cos-v1	EXAI	0.886	0.915	0.850	0.944	0.895	0.924	0.934
	MH	0.889	0.938	0.889	0.889	0.889	0.889	0.889
	LEARNSB	0.842	0.946	0.152	0.954	0.262	0.464	0.624
	average	0.872	0.933	0.630	0.929	0.682	0.759	0.816
deberta-v3-base	EXAI	0.943	0.925	1,000	0.889	0.941	0.909	0.899
	MH	0.806	0.926	0.867	0.722	0.788	0.747	0.734
	LEARNSB	0.762	0.926	0.109	0.989	0.196	0.378	0.547
	average	0.837	0.926	0.659	0.867	0.642	0.678	0.727

5 Results

This section presents the results of an extensive set of experiments of fine-tuning transformer-based models and ensembles for the task of scientific paper classification, evaluated in terms of several metrics. A total of 322 instances of fine-tuned Hugging Face models and 371 ensemble combinations were created. The results shown in this section represent the best performing models and ensembles, as the entire content of the results produced would have been too extensive for this paper.

5.1 Transformers

The results of the transformer evaluation were obtained through a process described in subsection 4.4.1, and presented in Table 3.

The results in Table 3, refer to five of the higher performing models (all-mpnet-base-v2, splade-cocondenser-ensembledistil, bert-base-uncased, multi-qa-mpnet-base-dot-v1, multi-qa-distilbert-cos-v1) in terms of recall and accuracy. One model with lower performance for comparison (deberta-v3-base) is also shown. The results are presented separately for each of the three datasets, along with their average. The average recall for the best models is above 0.9 and even above 0.85 for the deberta-v3-base. The average accuracy of the models reaches above 0.88 in the best cases and remains above 0.85 for the other best performing models. The AUC value is above 0.9 for all tests, showing that the models perform excellently in terms of overall classification ability.

Based on the results, it can be concluded that all models have poor precision for the LEARNBS dataset due to the highly unbalanced

evaluation set. It can be implied that the set of papers that the model classifies as relevant will contain a lot of irrelevant papers (false positives) for a highly unbalanced set of examples. Although it would be better to have high precision as well, this is not an issue as recall and accuracy are still high, assisting in retrieving relevant papers. The F-beta increases as beta increases, favoring the recall metric over precision. The F1 metric shows how good models perform in terms of precision and recall and overall, this metric achieves good results, except for the LEARNBS where the poor F1 score stands from poor precision values. In most experiments, as beta increases, the F-beta score also increases, showing that the recall value is high even in cases where the precision values are low. In cases where the F-beta metric drops with the increase of the beta value, the value does not drop by a large percentage and it is still considered good, which indicates that the recall value is excellent and stable regardless of the precision value.

5.2 Ensembles

Following the process described in 4.4.2, ensembles were evaluated, and the selected results are shown in Table 4.

The table shows the results obtained from the selected ensembles. The F1 and F2 metrics are not shown in this table, as in this case F3 is a more relevant metric, favoring recall over precision. In general, the results of the ensembles are relatively similar, with minor differences between them. Although the results of these ensembles are generally not as good as those of single models, ensembles formed from well-performing models achieve decent accuracy and

Table 4: Selected ensembles' metric evaluation results.

Base models in an ensemble	Dataset	Accuracy	AUC score	Precision	Recall	F3
1. deberta-v3-base	EXAI	0.914	0.915	0.941	0.889	0.894
2. all-mpnet-base-v2	MH	0.889	0.889	0.938	0.833	0.843
3. multi-qa-distilbert-cos-v1	LEARNsb	0.821	0.902	0.140	0.989	0.616
	average	0.875	0.902	0.673	0.904	0.784
1. all-mpnet-base-v2	EXAI	0.914	0.915	0.941	0.889	0.894
2.	MH	0.917	0.917	0.895	0.944	0.939
splade-cocondenser-ensembledistil	LEARNsb	0.827	0.894	0.142	0.966	0.612
3. bert-base-uncased	average	0.886	0.909	0.659	0.933	0.815
1. all-mpnet-base-v2	EXAI	0.914	0.915	0.941	0.889	0.894
2.	MH	0.917	0.917	0.895	0.944	0.939
splade-cocondenser-ensembledistil	LEARNsb	0.830	0.896	0.145	0.966	0.616
3. bert-base-uncased	average	0.887	0.909	0.660	0.933	0.816
4. multi-qa-distilbert-cos-v1						
1. albert/albert-base-v2	EXAI	0.914	0.915	0.941	0.889	0.894
2. deberta-v3-base	MH	0.889	0.889	0.889	0.889	0.889
3. all-mpnet-base-v2	LEARNsb	0.825	0.893	0.140	0.966	0.608
4.	average	0.876	0.899	0.657	0.914	0.683
splade-cocondenser-ensembledistil						
5. multi-qa-mpnet-base-dot-v1						

recall metrics. Comparing the performance of the first ensemble that contains deberta-v3-base with the performance of the individual deberta-v3-base model, the assumptions about the ensemble evaluation method were proven correct. For the EXAI dataset, on which deberta-v3-base performed excellently, other models in the ensemble had worse accuracy, AUC and precision so they reduced the model's performance on this dataset. Conversely, other models in the first ensemble improved the results on MH dataset, where the deberta-v3-base model had less favorable results. Therefore, ensemble evaluation offers a compromise between increasing the probability of obtaining satisfactory results and sacrificing excellent results that certain models achieve on specific datasets.

6 Conclusion

This research highlights the benefits of using transformer-based models to automate systematic literature reviews significantly streamlining a traditionally labor-intensive process. By experimenting with tools such as Hugging Face AutoTrain and various transformer models their ability to classify scientific text with high precision was demonstrated.

The Hugging Face AutoTrain tool enabled the development of accurate binary classification systems to identify relevant scientific papers. This success is due to extensive pretraining and fine-tuning that improves performance in NLP tasks, especially in understanding domain-specific terminology. The best models achieved accuracy higher than 0.88 and recall higher than 0.9 in three used datasets, demonstrating their effectiveness. The ensemble approach, which combines the predictions of multiple models, proved effective in achieving stable results, although it sometimes reduced the peak performance of the best individual models. These results show the importance of carefully selecting the ensemble components to achieve a balance between stability and performance.

The balance of the dataset emerged as a critical factor, especially for highly unbalanced datasets such as LEARN. A LEARNsb variant with balanced subsets, significantly improved recall, which is crucial for systematic reviews. This emphasizes the importance of preprocessing in machine learning tasks, especially in applications with unbalanced data.

Despite the promising results, this study identified several limitations, such as the high computational cost and the time for training and evaluation, which may be a limiting factor for smaller research teams or projects with limited computational capacity. Fine-tuning and optimization of hyperparameters, especially for large datasets, remains a resource-intensive challenge. Future research could explore more efficient training techniques or cloud-based to address these issues.

Future improvements could also include zero-shot learning for classifying new data without extensive fine-tuning to improve applicability to different topics and datasets: Research into new architectures or hybrid models could further improve performance, especially in interdisciplinary use. Overall, the research provides a solid foundation for AI-powered automation of literature reviews. It provides a practical tool for managing large volumes of scientific literature with high accuracy and recall. Although human supervision is still necessary, this approach has the potential to significantly reduce the workload and accelerate knowledge synthesis.

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